

RED REPORT

What's the evidence that gabapentin improves survival in glioblastoma?

- Gabapentin is *not* an NCCN “standard-of-care” therapy for glioblastoma with the goal of improving overall survival. In guidelines, gabapentin’s usual role in neuro-oncology is **symptom management** (e.g., neuropathic pain) rather than disease-modifying therapy. ([NCCN](#))
- That said, a recent retrospective, multi-institutional study (Mass General Brigham + UCSF; total n≈1,072) reported that **postoperative gabapentin use was associated with longer overall survival** compared with no gabapentin, after adjustment/propensity methods. The UCSF cohort reported **median OS ~20.8 vs 14.7 months** and a multivariable hazard ratio suggesting benefit. **This is association, not proof of causality** (non-randomized; confounding remains possible). ([Nature](#))
- There is also **active debate/caution in the literature** about whether current observational data are sufficient to justify routine off-label use for survival, and efforts are ongoing to strengthen causal inference and/or test prospectively. ([OUP Academic](#))

1) In this patient, is gabapentin indicated to improve overall survival, or symptom control only?

Best evidence-based framing:

- If the goal is “improve overall survival,” gabapentin should be viewed as *experimental/off-label at this time*, supported mainly by retrospective association and mechanistic plausibility—not definitive randomized evidence. ([Nature](#))
- If the goal is **symptom control** (neuropathic pain, allodynia, certain headache syndromes, etc.), gabapentin can be appropriate **when those symptoms exist**, but it should be started with clear dosing and monitoring.

In *this* specific scenario (ECOG 3, significant cognitive impairment, recent fall, fatigue, shuffling gait):

- Starting gabapentin *solely* for a **putative survival benefit** is a **high-risk/low-certainty trade** because the patient already has multiple features that make gabapentin’s common adverse effects (somnolence, dizziness, gait instability, cognitive slowing) especially consequential. The FDA label highlights **dizziness and somnolence** as common adverse reactions and reasons for discontinuation. ([FDA Access Data](#))
- A very reasonable stance is: **reserve gabapentin for a clear symptomatic indication** unless a neuro-oncologist believes the potential survival upside (uncertain) meaningfully outweighs the near-term safety risks in this frail context.

2) If gabapentin is believed to confer a survival benefit, what dosing regimen is appropriate, and how to weigh benefit vs risk?

What dosing was “typical” in the survival-association study?

In the Nature Communications paper, the **median total daily gabapentin dose in the UCSF cohort was ~600 mg/day (IQR 300–900 mg/day)**. ([Nature](#))

Important: that study does *not* establish an “optimal anticancer dose,” and it does not mean higher is better—only what was commonly observed among users.

A safer approach if you and the treating team still choose an off-label trial

Because this patient is **80, ECOG 3, cognitively impaired, and has fallen**, a “start low, go slow, with stop rules” strategy is the only defensible way to minimize harm:

Practical, safety-first titration (general medical practice; must be individualized):

- **Start:** 100 mg **at bedtime** (or even lower if very sensitive), not PRN and not “family discretion.”
- **Reassess in 2–3 days** for: daytime sedation, worsened confusion, dizziness, gait decline, orthostasis, and any near-falls.
- If tolerated and still pursuing the off-label goal, consider gradual increases to **100 mg BID → 100 mg TID** over 1–2+ weeks, *only if* function and safety remain stable.
- Be cautious about pushing toward the “observed median” of ~600 mg/day in this patient; the incremental fall risk may outweigh uncertain benefit.

Explicit risk–benefit weighing in this patient

Potential (uncertain) upside

- Observational data suggest a **survival association** and plausible mechanism (neuron–glioma interaction / TSP-1 axis), but it remains **unproven** and not yet a guideline-endorsed disease-modifying therapy. ([Nature](#))

Concrete, near-term downside (high probability / high impact here)

- **Sedation, dizziness, ataxia/unsteadiness, cognitive slowing**—all especially problematic with ECOG 3, baseline cognitive impairment, and a recent fall. ([FDA Access Data](#))
- **Elderly patients** and those on other CNS depressants are at higher risk of serious adverse effects, including **respiratory depression** in certain settings; FDA has issued warnings about this risk profile (especially when combined with other sedating meds). ([U.S. Food and Drug Administration](#))

- If the patient has **renal impairment** (common at 80), gabapentin can accumulate and worsen sedation/gait instability—dose adjustment becomes crucial.

“Stop rules” you can ask the treating team to formalize

If you proceed, it’s reasonable to document: **hold or discontinue** if any of the following occur after initiation/escalation:

- new/worsening daytime somnolence or delirium
- worsened gait, near-falls, or any fall
- meaningful decline in participation in PT/OT or ADLs attributable to sedation
- caregiver reports “he’s noticeably more confused” after dose changes

What I would push back on from the surgeon’s instructions

A prescription that leaves **dose/frequency to family discretion** is unsafe here. If gabapentin is used at all, it should be with:

- a **fixed schedule**
- a **written titration plan**
- **clear monitoring targets**
- a named clinician responsible for dose changes (ideally neuro-oncology)

Bottom line

1. **Gabapentin is not currently “indicated” as standard-of-care to improve overall survival in glioblastoma**; the best human data to date are **retrospective** and should be treated as **hypothesis-generating**. In this patient, given frailty and fall risk, it is most defensible to **reserve gabapentin for symptom control** unless neuro-oncology strongly supports an off-label trial. ([Nature](#))
2. If you still choose to trial it off-label, the most evidence-anchored reference point is that real-world users in the large retrospective study often received around **300–900 mg/day (median ~600 mg/day)**—but for *this* patient, a **much more conservative start (e.g., 100 mg qHS)** with slow titration and strict stop rules is the safer way to balance uncertain benefit against very real risks (sedation/falls). ([Nature](#))

If you want, I can draft a one-paragraph message the family could send to neuro-oncology that (1) neutrally summarizes the surgeon’s claim, (2) cites the key retrospective paper, and (3) asks for a clear go/no-go recommendation with a safe dosing/monitoring plan.